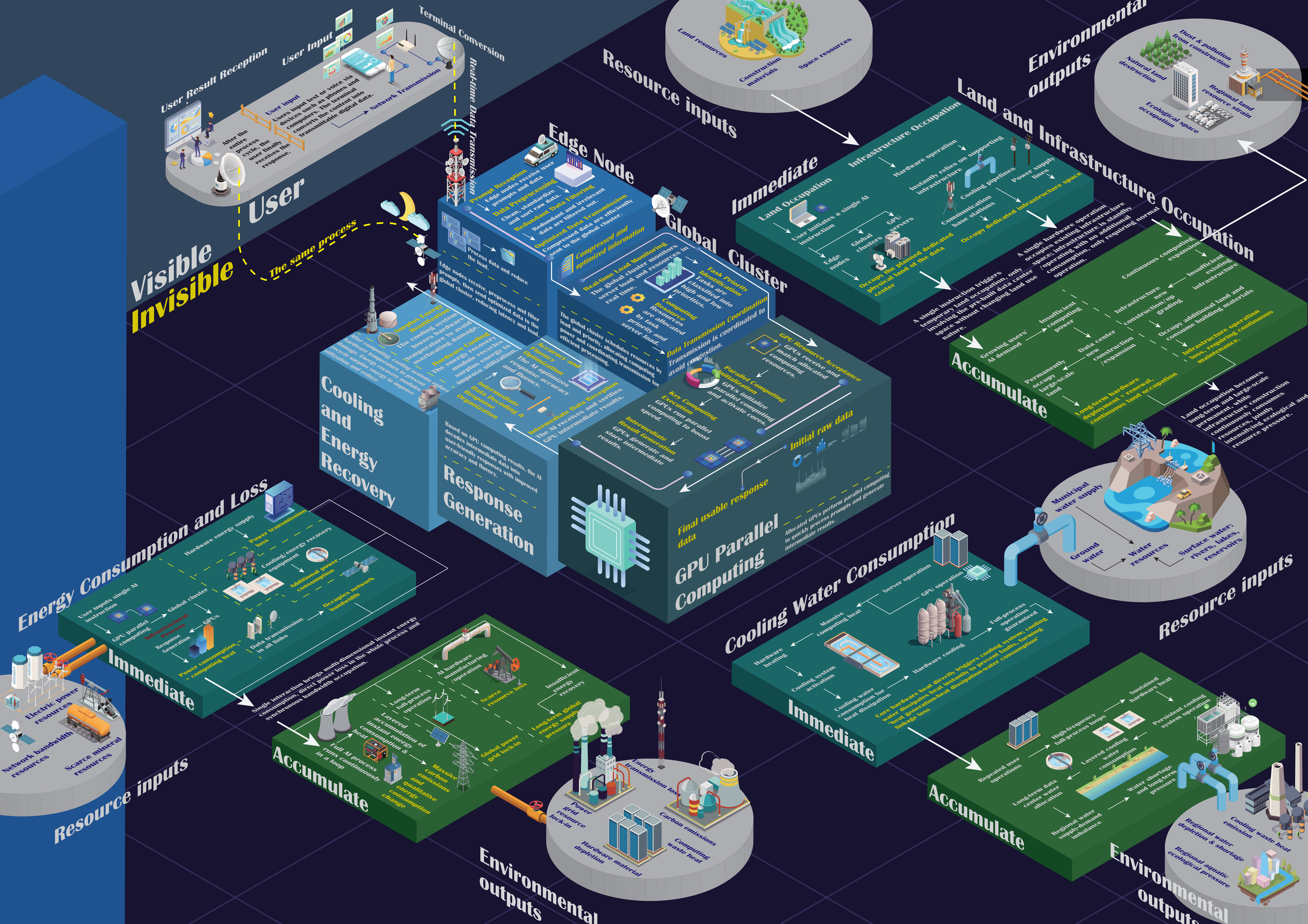
Visible
Invisible
The same process
User
User Result Reception
User Input
Terminal Conversion
Real-time Data Transmission
User input
Users input text or voice via devices such as phones and computers. The terminal converts the content into transmittable digital data.
Network Transmission
After the entire process cycle, the user finally receives the response.
Edge Node
Prompt Reception
Edge nodes receive user prompts and data
Data Preprocessing
Clean, standardize and sort raw data.
Redundant Data Filtering
Redundant and irrelevant data are filtered out.
Optimized Data Transmission
Compressed data are efficiently sent to the global cluster.
Compressed and optimized information
Process data and reduce the load.
Edge nodes receive, preprocess and filter prompts, then send optimized data to the global cluster, reducing latency and load.
Global Cluster
Real-time Load Monitoring
The global cluster monitors server load and resources in real time.
Task Priority Identification
Tasks are classified into high and low priorities.
Computing Resources
Resources are allocated by task priority and server load.
Data Transmission Coordination
Transmission is coordinated to avoid congestion.
The global cluster schedules resources by load and priority, allocating computing for power and coordinating transmission for efficient processing.
Cooling and Energy Recovery
Surplus Energy Reallocation
The cooling system towers hardware temperature to avoid overheating damage.
Hardware Cooling
The energy recovery module recycles surplus energy to improve utilization.
Response Generation
Accuracy and Fluency Optimization
The AI optimizes response accuracy and fluency.
Intermediate Data Reception
The AI receives and verifies GPU intermediate results.
Intermediate Data Decoding & Response Organization
Based on GPU computing results, the AI decodes intermediate data into user-friendly responses with improved accuracy and fluency.
GPU Parallel Computing
GPU Resource Acceptance
GPUs receive and match allocated computing resources.
Parallel Computing Initialization
GPUs initialize parallel computing and activate cores.
Key Computing Execution
GPUs run parallel computing to boost speed.
Intermediate Result Generation
GPUs generate and store intermediate results.
Initial raw data
Final usable response data
Allocated GPUs perform parallel computing to quickly process prompts and generate intermediate results.
Resource inputs
Land resources
Construction materials
Space resources
Immediate
Land Occupation
User initiates a single AI instruction
Edge nodes
Global clusters
GPU servers
Occupy the planned dedicated physical land of the data center
Infrastructure Occupation
Hardware operation
Instantly relies on supporting infrastructure
Cooling pipelines
Power supply lines
Communication base stations
Occupy dedicated infrastructure space
A single hardware operation occupies existing infrastructure space; infrastructure is standby /operating with no additional consumption, only requiring normal operation.
A single instruction triggers temporary land occupation, only involving the pre-built data center space without changing land use nature.
Accumulate
Growing users' AI demand
Insufficient computing power
Data center new construction /expansion
Permanently occupy large-scale land
Long-term hardware deployment / renewal, continuous land occupation
Infrastructure new construction/up grading
Continuous computing expansion
Insufficient existing infrastructure
Occupy additional land and consume building materials
Infrastructure operation loss, requiring continuous maintenance.
Land occupation becomes long-term and large-scale permanent, while infrastructure construction continuously consumes resources, jointly intensifying ecological and resource pressure.
Environmental outputs
Natural land destruction
Dust & pollution from construction
Regional land resource strain
Ecological space occupation
Energy Consumption and Loss
Immediate
User inputs single AI instruction
GPU parallel computing
Global cluster
Hardware energy supply
Power transmission loss
Cooling/energy recovery equipment
Additional power consumption
Occupies network bandwidth
Response Generation
GPUs
Data transmission in all links
Power consumption + computing heat
Single interaction brings multi-dimensional instant energy consumption: direct power loss in the whole process and synchronous bandwidth occupation.
Accumulate
Full AI process runs continuously in a loop
Heat
Layered accumulation of instant energy consumption & heat
Long-term full-process operation
AI hardware manufacturing /operation
Massive carbon emissions and qualitative energy consumption change
Scarce resource loss
Global power grid lock-in
Insufficient energy recovery
Long-term global energy supply pressure
Resource inputs
Electric power resources
Network bandwidth resources
Scarce mineral resources
Environmental outputs
Power grid resource lock-in
Energy transmission loss
Hardware material depletion
Carbon emissions
Computing waste heat
Cooling Water Consumption
Immediate
Server operation
GPU operation
Massive computing heat
Hardware heating
Cooling system activation
Cooling water consumption for heat dissipation
Hardware cooling
Full-process operation guaranteed
Core hardware heat directly triggers cooling system; cooling water dissipates heat instantly to prevent faults, forming heat generation-heat dissipation-water consumption linkage loss
Accumulate
Repeated user operations
High-frequency full-process loops
Sustained hardware heat
Persistent cooling system operation
Layered cooling water consumption
Long-term data center water allocation
Water shortage and long-term pressure
Regional water supply-demand imbalance
Resource inputs
Municipal water supply
Ground water
Water resources
Surface water: rivers, lakes, reservoirs
Environmental outputs
Cooling waste heat emission
Regional water depletion & shortage
Regional aquatic ecological pressure